

# Deep Generative Models

## Lecture 11

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AI Masters

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## Recap of Previous Lecture

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{f}_\theta(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0$$

### Theorem (Continuity Equation)

If  $\mathbf{f}$  is uniformly Lipschitz continuous in  $\mathbf{x}$  and continuous in  $t$ , then

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left( \frac{\partial \mathbf{f}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right)$$

### Solution of the Continuity Equation

$$\log p_1(\mathbf{x}(1)) = \log p_0(\mathbf{x}(0)) - \int_0^1 \text{tr} \left( \frac{\partial \mathbf{f}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right) dt.$$

- ▶ This solution gives us the density along the trajectory (not the total probability path).
- ▶ However, it's difficult to efficiently estimate **the last term**.

# Recap of Previous Lecture

## SDE Basics

Let's define a stochastic process  $\mathbf{x}(t)$  with initial condition  $\mathbf{x}(0) \sim p_0(\mathbf{x})$ :

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w},$$

where  $\mathbf{w}(t)$  is the standard Wiener process (Brownian motion):

$$\mathbf{w}(t) - \mathbf{w}(s) \sim \mathcal{N}(0, (t-s)\mathbf{I}), \quad d\mathbf{w} = \epsilon \cdot \sqrt{dt}, \text{ where } \epsilon \sim \mathcal{N}(0, \mathbf{I}).$$

## Discretization of SDE (Euler Method) - SDEsolve

$$\mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}(t), t) \cdot dt + g(t) \cdot \epsilon \cdot \sqrt{dt}$$

- ▶ At each time  $t$ , we have the density  $p_t(\mathbf{x}) = p(\mathbf{x}, t)$ .
- ▶  $p : \mathbb{R}^m \times [0, 1] \rightarrow \mathbb{R}_+$  is a **probability path** between  $p_0(\mathbf{x})$  and  $p_1(\mathbf{x})$ .

# Recap of Previous Lecture

## Theorem (Kolmogorov-Fokker-Planck)

The evolution of the distribution  $p_t(\mathbf{x})$  is given by:

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}g^2(t)\Delta_{\mathbf{x}}p_t(\mathbf{x})$$

## Langevin SDE (Special Case)

$$d\mathbf{x} = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) dt + \mathbf{1} \cdot d\mathbf{w}$$

The density  $p_{\theta}(\mathbf{x})$  is a **stationary** distribution for the SDE.

## Langevin Dynamics

Samples from the following dynamics will come from  $p_{\theta}(\mathbf{x})$  under mild regularity conditions for a small enough  $\eta$ :

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \frac{\eta}{2} \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \sqrt{\eta} \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}).$$

# Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

## Variance Exploding SDE (NCSN)

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}, \quad \mathbf{f}(\mathbf{x}, t) = 0, \quad g(t) = \sqrt{\frac{d[\sigma^2(t)]}{dt}}$$

The variance grows since  $\sigma(t)$  is a monotonically increasing function.

## Variance Preserving SDE (DDPM)

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}(t), \quad g(t) = \sqrt{\beta(t)}$$

The variance is preserved if  $\mathbf{x}(0)$  has unit variance.

*Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020*

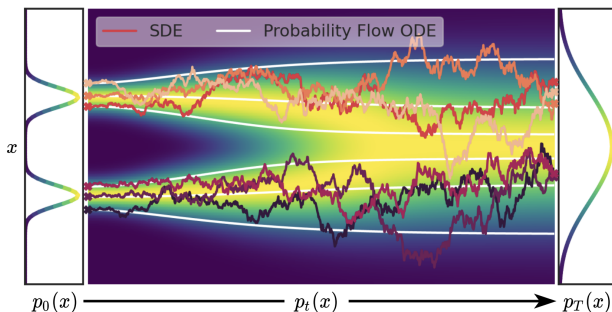
# Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \quad (\text{SDE with the probability path } p_t(\mathbf{x}))$$

## Probability Flow ODE

There exists an ODE with the identical probability path  $p_t(\mathbf{x})$  of the form:

$$d\mathbf{x} = \left( \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$



Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

# Outline

1. Reverse SDE
2. Score-Based Generative Models Through SDEs
3. Flow Matching
4. Conditional Flow Matching

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## Reverse SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt, \quad \mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}, t)dt$$

Here  $dt$  can be  $> 0$  or  $< 0$ .

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## Reverse ODE

Let  $\tau = 1 - t$  ( $d\tau = -dt$ ).

$$d\mathbf{x} = -\mathbf{f}(\mathbf{x}, 1 - \tau)d\tau$$

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- ▶ How do we reverse the SDE  $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ ?
- ▶ The Wiener process introduces randomness that must be reversed.

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## Theorem

There exists a reverse SDE for  $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ , given by:

$$d\mathbf{x} = \left( \mathbf{f}(\mathbf{x}, t) - g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt + g(t)d\bar{\mathbf{w}}$$

where  $dt < 0$  and  $\bar{\mathbf{w}}$  is a reverse-time Wiener process.

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where  $dt < 0$  and  $\bar{\mathbf{w}}(t)$  is a reverse-time Wiener process.

**Note:** Again, the score function appears:  $\mathbf{s}(\mathbf{x}, t) = \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$ .

## Proof Sketch

- ▶ Convert the initial SDE to a probability flow ODE.
- ▶ Reverse the probability flow ODE.
- ▶ Convert the reversed probability flow ODE back to an SDE.

# Proof of Reverse SDE

- Convert the initial SDE to a probability flow ODE:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$d\mathbf{x} = \left( \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$

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- Reverse the probability flow ODE:

$$d\mathbf{x} = \left( \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$

$$d\mathbf{x} = \left( -\mathbf{f}(\mathbf{x}, 1 - \tau) + \frac{1}{2}g^2(1 - \tau)\frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x}) \right) d\tau$$



# Proof of Reverse SDE

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- Convert the reversed probability flow ODE back to an SDE:

$$d\mathbf{x} = \left( -\mathbf{f}(\mathbf{x}, 1 - \tau) + \frac{1}{2}g^2(1 - \tau)\frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x}) \right) d\tau$$

$$d\mathbf{x} = \left( -\mathbf{f}(\mathbf{x}, 1 - \tau) + g^2(1 - \tau)\frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x}) \right) d\tau + g(1 - \tau)d\mathbf{w}$$

Here  $\mathbf{w}(\tau)$  is a Wiener process in forward  $\tau$ -time.

# Reverse SDE

## Theorem

There exists a reverse SDE for  $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ , given by:

$$d\mathbf{x} = \left( \mathbf{f}(\mathbf{x}, t) - g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt + g(t) d\bar{\mathbf{w}}$$

where  $dt < 0$  and  $\bar{\mathbf{w}}(t)$  is a reverse-time Wiener process.

## Proof (Continued)

$$d\mathbf{x} = \left( -\mathbf{f}(\mathbf{x}, 1 - \tau) + g^2(1 - \tau) \frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x}) \right) d\tau + g(1 - \tau) d\mathbf{w}$$

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Here  $d\tau > 0$ ,  $dt < 0$ , and  $\bar{\mathbf{w}}$  is reverse-time Wiener process.

## Reverse SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \quad - \text{SDE}$$

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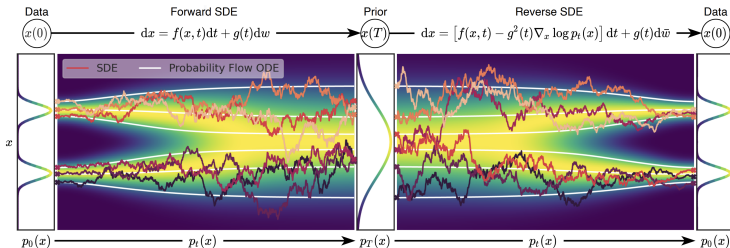
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- ▶ This framework allows us to transform one distribution into another via an SDE with a prescribed probability path  $p_t(\mathbf{x})$ .
- ▶ We can invert this process using the score function.



Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

## Score Matching SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

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$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}, \quad \mathbf{f}(\mathbf{x}, t) = 0, \quad g(t) = \sqrt{\frac{d[\sigma^2(t)]}{dt}}$$



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$$d\mathbf{x} = \left( -\frac{1}{2} \frac{d[\sigma^2(t)]}{dt} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt \quad (\text{probability flow ODE})$$

$$d\mathbf{x} = \left( -\frac{d[\sigma^2(t)]}{dt} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt + \sqrt{\frac{d[\sigma^2(t)]}{dt}} d\bar{\mathbf{w}} \quad (\text{reverse SDE})$$

# Diffusion SDE

## Variance Preserving SDE

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}(t), \quad g(t) = \sqrt{\beta(t)}$$

Variance is preserved as long as  $\mathbf{x}(0)$  has unit variance.

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# Outline

1. Reverse SDE
2. Score-Based Generative Models Through SDEs
3. Flow Matching
4. Conditional Flow Matching

# Score-Based Generative Models Through SDEs

## Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \mathbf{s}_{\theta, t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2$$

Is it possible to train score-based diffusion models in continuous time?

# Score-Based Generative Models Through SDEs

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Is it possible to train score-based diffusion models in continuous time?

## Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0, 1]} \mathbb{E}_{q(\mathbf{x}(t) | \mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t) | \mathbf{x}(0)) \right\|_2^2$$

# Score-Based Generative Models Through SDEs

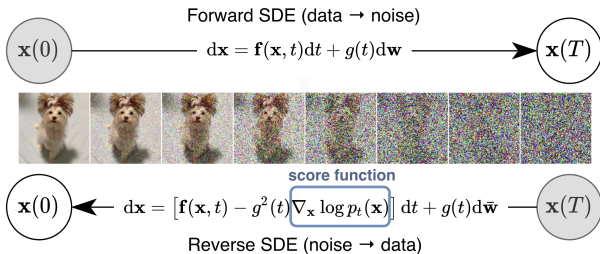
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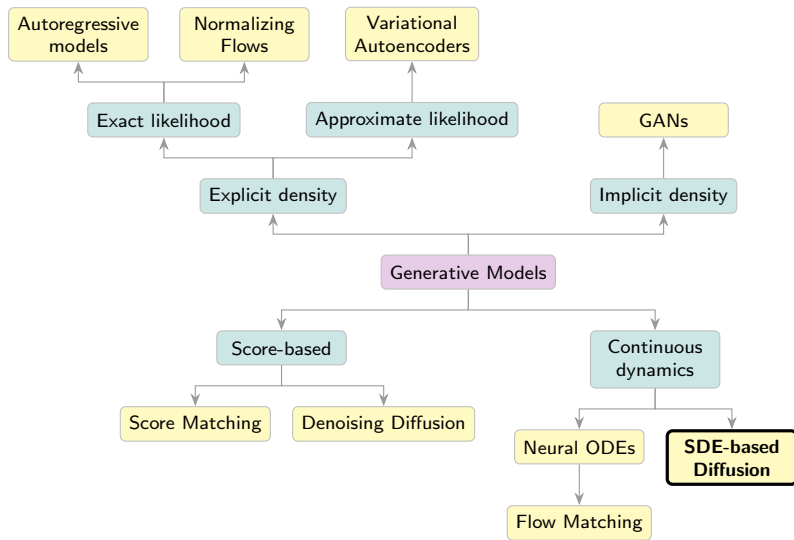
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Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

# Generative Models Taxonomy





# Score-Based Generative Models Through SDEs

## Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

# Score-Based Generative Models Through SDEs

## Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(\mathbf{x}(0), t), \boldsymbol{\sigma}^2(t, \mathbf{x}(0)) \cdot \mathbf{I}\right)$$

$$\nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) = -\frac{1}{\boldsymbol{\sigma}^2} \odot (\mathbf{x}(t) - \boldsymbol{\mu})$$

**Note:** Normality holds for  $\mathbf{f}(\mathbf{x}, t)$  affine in  $\mathbf{x}$  ( $\mathbf{f}(\mathbf{x}, t) = f(t)\mathbf{x}$ ).

# Score-Based Generative Models Through SDEs

## Continuous-Time Objective

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$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w} \quad (\text{Variance Exploding SDE})$$

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w} \quad (\text{Variance Preserving SDE})$$

Is it possible to explicitly derive  $\boldsymbol{\mu}(\mathbf{x}(0), t)$  and  $\boldsymbol{\Sigma}(\mathbf{x}(0), t)$  for VE-SDE and VP-SDE?

# Score-Based Generative Models Through SDEs

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(\mathbf{x}(0), t), \boldsymbol{\sigma}^2(t, \mathbf{x}(0)) \cdot \mathbf{I}\right)$$

## Theorem

The moments of the SDE  $d\mathbf{x} = f(t)\mathbf{x}dt + g(t)d\mathbf{w}$  satisfy:

$$\frac{d\boldsymbol{\mu}(\mathbf{x}(0), t)}{dt} = f(t)\boldsymbol{\mu}(\mathbf{x}(0), t)$$

$$\frac{d\boldsymbol{\sigma}^2(t, \mathbf{x}(0))}{dt} = 2f(t)\boldsymbol{\sigma}^2(t, \mathbf{x}(0)) + g^2(t) \cdot \mathbf{I}$$

# Score-Based Generative Models Through SDEs

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(\mathbf{x}(0), t), \boldsymbol{\sigma}^2(t, \mathbf{x}(0)) \cdot \mathbf{I}\right)$$

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## Proof

$$\mathbb{E}[d\mathbf{x}|\mathbf{x}(0)] = \mathbb{E}[f(t)\mathbf{x}dt|\mathbf{x}(0)] + \mathbb{E}[g(t)d\mathbf{w}|\mathbf{x}(0)]$$

# Score-Based Generative Models Through SDEs

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(\mathbf{x}(0), t), \boldsymbol{\sigma}^2(t, \mathbf{x}(0)) \cdot \mathbf{I}\right)$$

## Theorem

The moments of the SDE  $d\mathbf{x} = f(t)\mathbf{x}dt + g(t)d\mathbf{w}$  satisfy:

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# Score-Based Generative Models Through SDEs

## Theorem

$$\frac{d\mu(\mathbf{x}(0), t)}{dt} = f(t)\mu(\mathbf{x}(0), t)$$



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# Score-Based Generative Models Through SDEs

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$$\mathbb{E}[d\mathbf{x}|\mathbf{x}(0)] = f(t)\mu(\mathbf{x}(0), t)dt$$

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$$\mu = \mathbf{x}(0) \exp\left(-\frac{1}{2} \int_0^t \beta(s) ds\right)$$

# Score-Based Generative Models Through SDEs

## Training

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(\mathbf{x}(0), t), \boldsymbol{\Sigma}(\mathbf{x}(0), t)\right)$$

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# Score-Based Generative Models Through SDEs

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$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0)e^{-\frac{1}{2} \int_0^t \beta(s) ds}, \left(1 - e^{-\int_0^t \beta(s) ds}\right) \cdot \mathbf{I}\right)$$

Here we omit the derivations of the variance.

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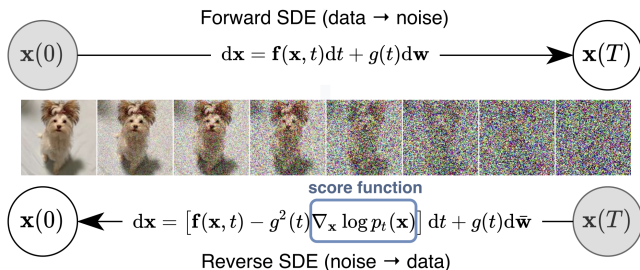
*Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020*



# Score-Based Generative Models Through SDEs

## Sampling

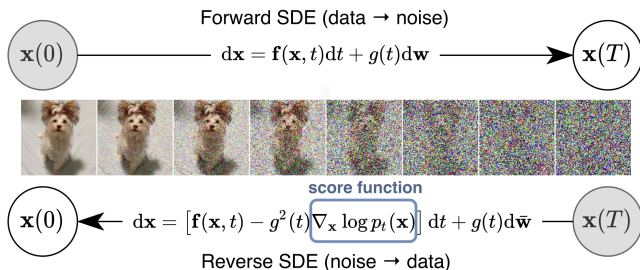
Solve the reverse SDE using numerical solvers (SDEsolve).



# Score-Based Generative Models Through SDEs

## Sampling

Solve the reverse SDE using numerical solvers (SDEsolve).



- ▶ Discretizing the reverse SDE provides ancestral sampling.
- ▶ Discretizing the probability flow ODE yields deterministic sampling.

# Outline

1. Reverse SDE
2. Score-Based Generative Models Through SDEs
3. Flow Matching
4. Conditional Flow Matching

# Continuous-Time Normalizing Flows

Let's return to ODE dynamics  $\mathbf{x}_t = \mathbf{x}(t)$  in the interval  $t \in [0, 1]$ :

- ▶  $\mathbf{x}_0 \sim p_0(\mathbf{x}) = p(\mathbf{x})$ ,  $\mathbf{x}_1 \sim p_1(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$ ;
- ▶  $p(\mathbf{x})$  is a base distribution (e.g.,  $\mathcal{N}(0, \mathbf{I})$ ), and  $p_{\text{data}}(\mathbf{x})$  is the true data distribution.

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{f}(\mathbf{x}_t, t), \quad \text{with initial condition } \mathbf{x}(0) = \mathbf{x}_0.$$

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## KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\text{div}(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left( \frac{\partial \mathbf{f}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right)$$

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- ▶ It's hard to solve the continuity equation directly due to the trace term.
- ▶ There's a method (the adjoint method) that solves this equation directly, but it's unstable and unscalable.

# Continuous-Time Normalizing Flows

## KFP Theorem (Continuity Equation)

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- ▶ Knowing the vector field  $\mathbf{f}(\mathbf{x}, t)$ , the KFP (or continuity) equation allows us to compute the density  $p_t(\mathbf{x})$ .
- ▶ Flow matching provides an alternative approach to Neural ODEs.

# Continuous-Time Normalizing Flows

## KFP Theorem (Continuity Equation)

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## Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{f}(\mathbf{x}, t) - \mathbf{f}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$



# Continuous-Time Normalizing Flows

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- ▶ Knowing the vector field  $\mathbf{f}(\mathbf{x}, t)$ , the KFP (or continuity) equation allows us to compute the density  $p_t(\mathbf{x})$ .
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## Flow Matching

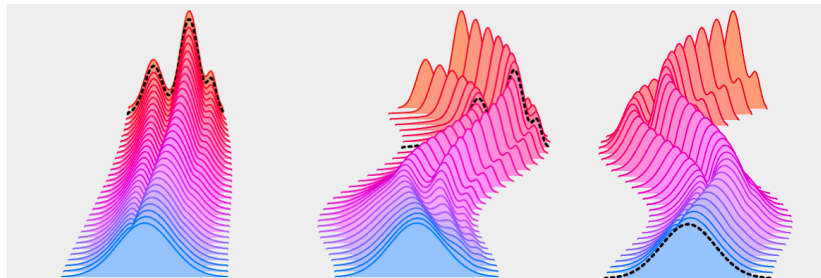
$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{f}(\mathbf{x}, t) - \mathbf{f}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

- ▶ Approximate the true vector field  $\mathbf{f}(\mathbf{x}, t)$  using  $\mathbf{f}_\theta(\mathbf{x}, t)$ .
- ▶ Use  $\mathbf{f}_\theta(\mathbf{x}, t)$  for deterministic sampling from the ODE.

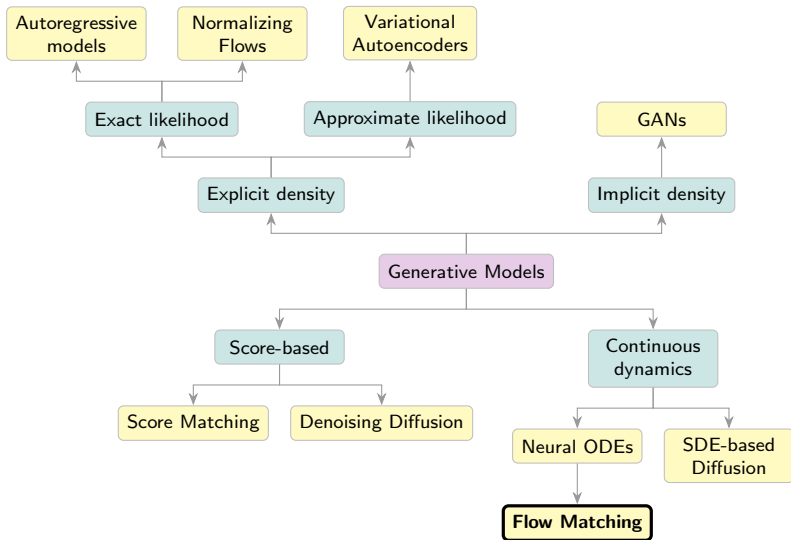
# Flow Matching

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- ▶ There are infinitely many possible  $\mathbf{f}(\mathbf{x}, t)$  between  $p_{\text{data}}(\mathbf{x})$  and  $p(\mathbf{x})$ .
- ▶ The true vector field  $\mathbf{f}(\mathbf{x}, t)$  is **unknown**.
- ▶ We need to select the "best"  $\mathbf{f}(\mathbf{x}, t)$  and make the objective tractable.



# Generative Models Taxonomy



# Outline

1. Reverse SDE
2. Score-Based Generative Models Through SDEs
3. Flow Matching
4. Conditional Flow Matching

# Flow Matching

Let's introduce the latent variable  $\mathbf{z}$ :

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Here,  $p_t(\mathbf{x}|\mathbf{z})$  is a **conditional probability path**.

# Flow Matching

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Here,  $p_t(\mathbf{x}|\mathbf{z})$  is a **conditional probability path**. The conditional probability path  $p_t(\mathbf{x}|\mathbf{z})$  satisfies the KFP theorem:

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}(\mathbf{f}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})),$$

where  $\mathbf{f}(\mathbf{x}, \mathbf{z}, t)$  is a **conditional vector field**:

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$$\frac{d\mathbf{x}_t}{dt} = \mathbf{f}(\mathbf{x}_t, t) \quad \Rightarrow \quad \frac{d\mathbf{x}_t}{dt} = \mathbf{f}(\mathbf{x}_t, \mathbf{z}, t)$$

What's the relationship between  $\mathbf{f}(\mathbf{x}, t)$  and  $\mathbf{f}(\mathbf{x}, \mathbf{z}, t)$ ?

# Flow Matching

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}(\mathbf{f}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})),$$

## Theorem

The following vector field generates the probability path  $p_t(\mathbf{x})$ :

$$\mathbf{f}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})} \mathbf{f}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{f}(\mathbf{x}, \mathbf{z}, t) \frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})} d\mathbf{z}$$



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## Proof

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# Flow Matching

## Flow Matching (FM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{f}(\mathbf{x}, t) - \mathbf{f}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

## Conditional Flow Matching (CFM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{f}(\mathbf{x}, \mathbf{z}, t) - \mathbf{f}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

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## Theorem

If  $\text{supp}(p_t(\mathbf{x})) = \mathbb{R}^m$ , then the optimal value of the FM objective equals the optimal value of the CFM objective.

# Flow Matching

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## Proof

This can be proved in a similar way as in the denoising score matching theorem.



# Summary

- ▶ SDEs can be reversed using the score function.
- ▶ Score matching (NCSN) and diffusion models (DDPM) are discretizations of SDEs (variance exploding and variance preserving).
- ▶ It's possible to train continuous-in-time score-based generative models using forward and reverse SDEs.
- ▶ Discretizing the reverse SDE yields ancestral sampling of the DDPM.
- ▶ Flow matching suggests fitting the vector field directly.
- ▶ Conditional flow matching introduces the latent variable  $\mathbf{z}$ , reformulating the initial task in terms of conditional dynamics.